

EMCode Quick-Start Guide

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EMCode is an integrated coding environment tool with auto-grading that will be used to grade the course coding assignments.

How do I open an EMCode assignment?

Whenever you are required to complete a programming assignment, you will see a Canvas page with a similar structure to the one below.

Programming Assignment 2.1: Multilayer Perceptron (Assignment 1A) [10 hours]

Due

No due date

Points

100

Submitting

an external tool



@ Learning outcome addressed

13. Implement the forward pass, backward pass, and stochastic gradient descent for a multilayer perceptron.

Now that you've explored the structure, function, and training of multilayer perceptron, you have an opportunity to create and train your own neural network to demonstrate your understanding of the methods.

Additional Details

Follow the directions in the programming assignment itself.

Submission instructions

Refer to the [Quick-Start Guide](#) on submitting the assignment.

The average completion time for this activity is 5–10 hours.

This is a required assignment and counts toward program completion.

Associated readings

For additional details, please visit the [Reading List](#) for suggestions.

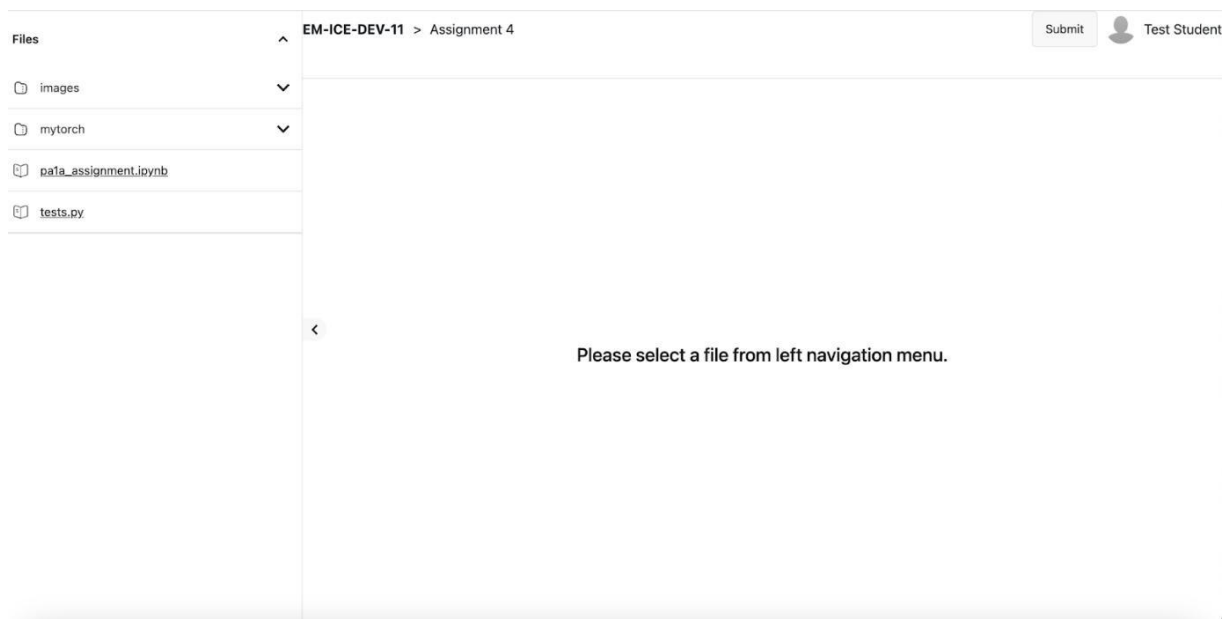
This tool needs to be loaded in a new browser window

Load Programming Assignment 2.1: Multilayer Perceptron (Assignment 1A) [10 hours] in a new window

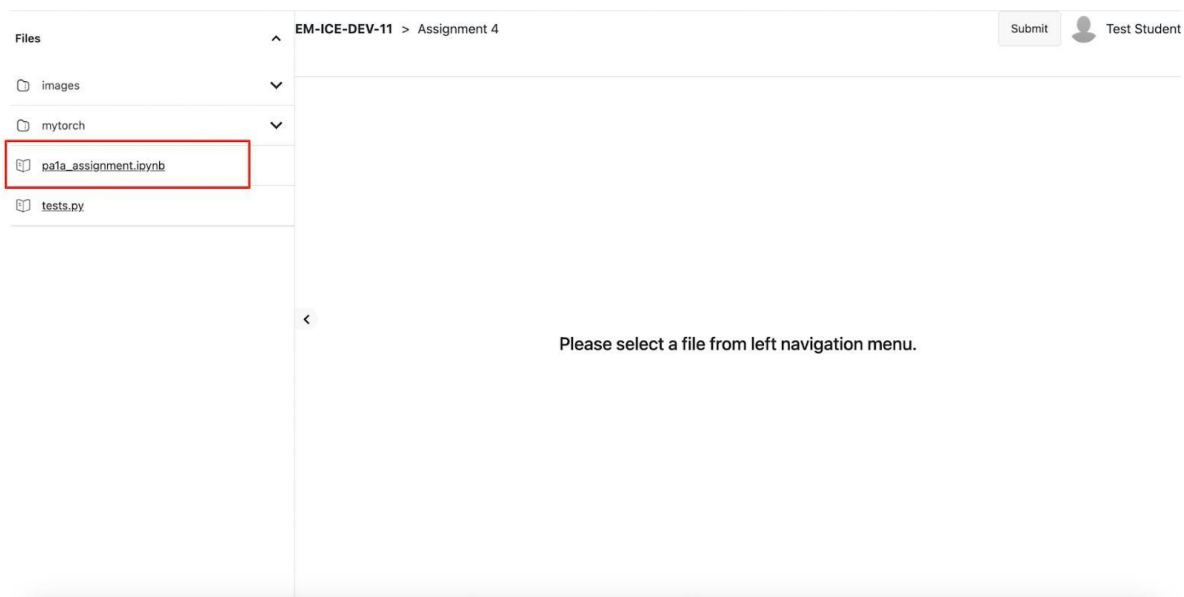
By selecting the link at the bottom of the page, you will be redirected to the EMMCode assignment page.

Note:

- It is recommended to access the EMMCode platform using the most recent version of Chrome, Safari, Firefox or Edge browsers.
- The EMMCode platform can **take up to 40 seconds to load** the first time you use it. Upon loading, you will see a screen similar to this:



Navigate to the .ipynb file of the assignment and click on it on the left panel:



The assignment will load in a Jupyter Notebook as shown below.

EM-ICE-DEV-11 > Assignment 4

Submit Test Student

pa1a_assignment.ipynb

Jupyter pa1a_assignment Last Checkpoint: 1 hour ago

File Edit View Run Kernel Settings Help

JupyterLab Python 3 (ipykernel)

Assignment 1 Part A - Multi-Layer Perceptrons (MLPs)

Welcome to the first assignment!

You'll be implementing your own basic version of PyTorch (that we'll cleverly call `mytorch`), using nothing but NumPy.

Introduction

Let's begin with a high-level recap on how NNs are trained.

Modern NNs are generally trained by repeating these three steps:

- 1. Forward Propagation**
 - a. Provide batch of observations to network, network tries to guess label of each observation by estimating the likelihood of each possible label
 - b. Loss function "scores" guesses by comparing them to the true label and calculating the average loss value across the batch.

The diagram illustrates the forward propagation process: **Batched Observations** are fed into a **Network** (comprising **Linear**, **ReLU**, and **Linear** layers). The network outputs a **Batch of guessed label likelihoods**, which are then processed by a **Loss Function** (CrossEntropyLoss) to yield a final **Loss = 2.342**.

Read the directions carefully and proceed with the assignment. Sometimes, the assignment may contain tips or important information on how to complete the assignment.

How do I use Jupyter Notebook?

Scrolling through the assignment, you will see EMMCode code cells (1) containing Python code. It's important that you run all the cells in proper sequence and to make sure your code is getting executed and the submission works correctly.

To run a code cell, select the cell. Then, select the "Run" icon (2) at the top or use the shortcut shift+return or shift+enter on your keyboard.

EM-ICE-DEV-11 > Assignment 4

Submit Test Student

pa1a_assignment.ipynb

Jupyter pa1a_assignment Last Checkpoint: 1 hour ago

File Edit View Run Kernel Settings Help

JupyterLab Python 3 (ipykernel)

Not Trusted

- Make sure to run the cell above where we import everything.
- After you complete your code, run the cell below to check that your implementation was correct.

1

```
def test_linear_forward_1(Linear):
    """[Given] Just to demonstrate how you'll be graded.

    Args:
        Linear (class): the entire class imported from 'nn.py'

    Returns:
        np.array: the output of passing through your Linear.forward()
    """
    # Initialize layer that feeds 3 input channels to 4 neurons.
    layer = Linear(3, 4)
    # Weights/biases are normally initialized randomly to small floats centered around 0,
    # but we'll manually set them like this for consistency/interpretability
    layer.weight = np.array([[1., 2., 3., 4.],
                             [5., 6., 7., 8.],
                             [9., 10., 11., 12.]])
    layer.bias = np.array([1., 2., 3., 4.])

    # Input array shaped (batch_size=2, in_features=3)
    x = np.array([[1., 2., 3.],
                  [4., 5., 6.]])

    # Run the input through Linear.forward().
    out = layer.forward(x)
    return out
```

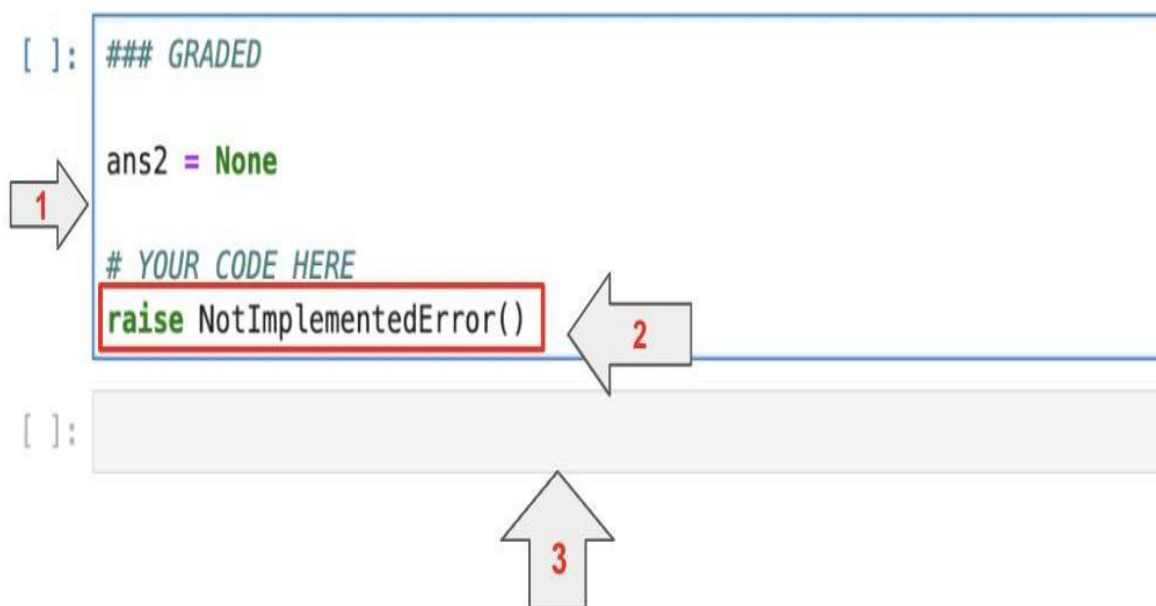
After doing this, a number will appear inside the square brackets to the left of the code cell. This number indicates that the particular code cell has executed successfully.

How do I answer a question?

Throughout the assignment, you will see various questions. In some cases, the questions might be connected. Therefore, it's important that you solve all the questions consecutively. Remember to take advantage of the support available to you if you get stuck. You can always pause your work to submit a support ticket or to post a question in Canvas. Each question will have two or more code cells below the question prompt. Additionally, you may be provided with hints or auxiliary code for reference to help solve the question.

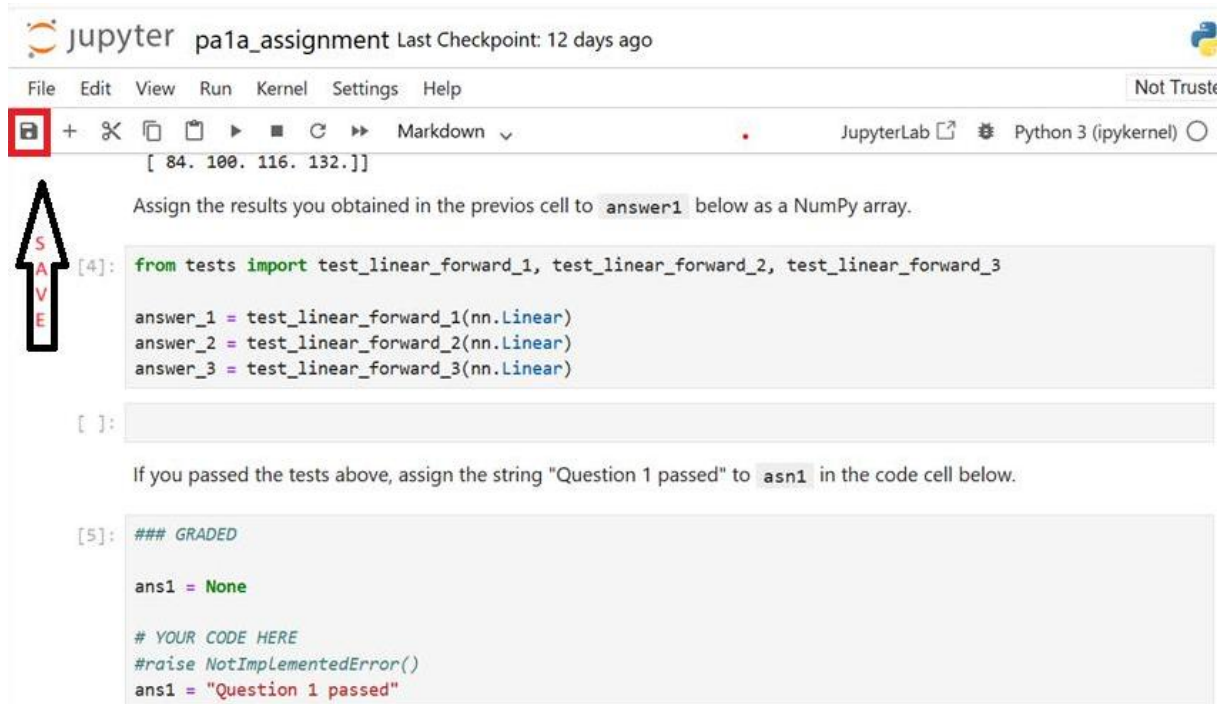
The first cell is a graded cell (1), and it contains auxiliary code for you to answer the question. Each question will have one or more code cells below the question prompt. You are required to enter their code in the graded cell, in the space provided “#YOUR CODE” . To write your solution, erase the line “raise NotImplementedError()” or comment this line and type your solution. Run the graded cell.

Below the graded cell, you will see one or more blank code cells (3). These are auto-grader cells that hide the solution to the question. To make sure that EMMCode can compare your solution against the actual one, be sure to run the auto-grader cells as well.



How do I submit an assignment?

To submit an assignment, first save your file by selecting the save icon. Be sure to save your assignment if you want to come back to it and finish it later.



The screenshot shows a JupyterLab window titled "jupyter pa1a_assignment Last Checkpoint: 12 days ago". The top menu bar includes File, Edit, View, Run, Kernel, Settings, and Help. A toolbar below the menu contains various icons, including a save icon (a floppy disk) which is highlighted with a red box. A black arrow points from the word "SAVE" (written vertically) to the save icon. The main area displays a code cell with the following text:

```
[4]: from tests import test_linear_forward_1, test_linear_forward_2, test_linear_forward_3

answer_1 = test_linear_forward_1(nn.Linear)
answer_2 = test_linear_forward_2(nn.Linear)
answer_3 = test_linear_forward_3(nn.Linear)
```

Below the code cell, there is a text prompt: "Assign the results you obtained in the previous cell to `answer1` below as a NumPy array."

Below that, another code cell is shown with the following text:

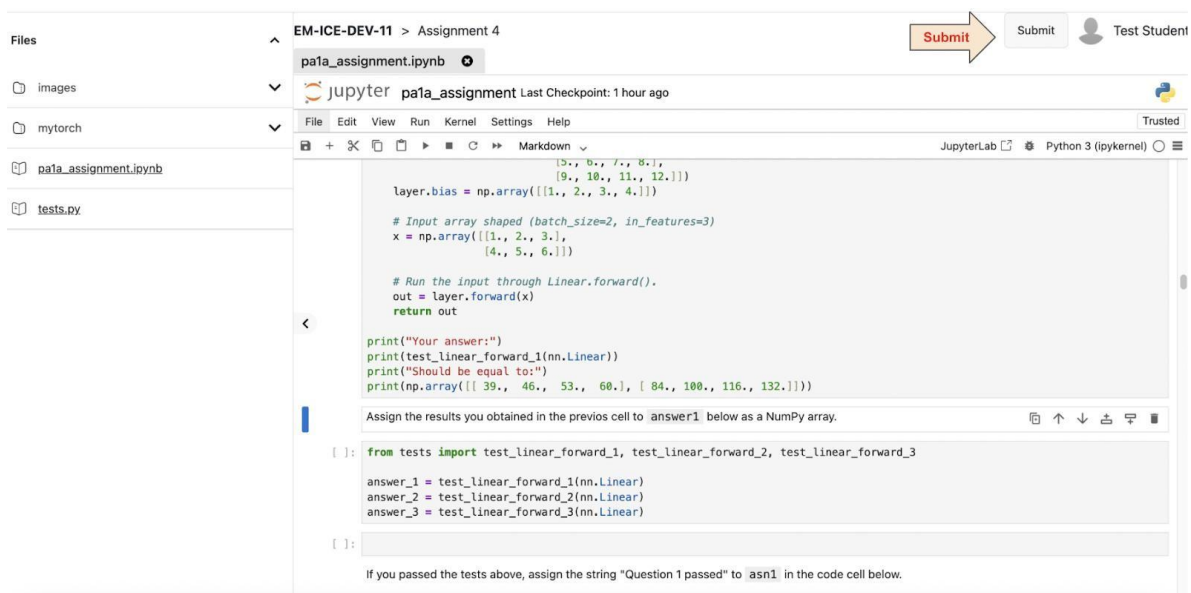
```
[5]: ### GRADED

ans1 = None

# YOUR CODE HERE
#raise NotImplementedError()
ans1 = "Question 1 passed"
```

Below the code cell, there is a text prompt: "If you passed the tests above, assign the string 'Question 1 passed' to `asn1` in the code cell below."

Then, select the "Submit" button on the top right.



The screenshot shows a JupyterLab window titled "EM-ICE-DEV-11 > Assignment 4". The top menu bar includes File, Edit, View, Run, Kernel, Settings, and Help. A toolbar below the menu contains various icons, including a submit icon (a document with an arrow) which is highlighted with a red box. A black arrow points from the word "Submit" (written vertically) to the submit icon. The main area displays a code cell with the following text:

```
[5]: from tests import test_linear_forward_1, test_linear_forward_2, test_linear_forward_3

answer_1 = test_linear_forward_1(nn.Linear)
answer_2 = test_linear_forward_2(nn.Linear)
answer_3 = test_linear_forward_3(nn.Linear)
```

Below the code cell, there is a text prompt: "Assign the results you obtained in the previous cell to `answer1` below as a NumPy array."

Below that, another code cell is shown with the following text:

```
[5]: ### GRADED

ans1 = None

# YOUR CODE HERE
#raise NotImplementedError()
ans1 = "Question 1 passed"
```

Below the code cell, there is a text prompt: "If you passed the tests above, assign the string 'Question 1 passed' to `asn1` in the code cell below."

At this point, EMMCode will start grading your assignment, and you should be able to see your results shortly after.

Each assignment is designed to allow multiple attempts before the due date. You can resubmit the assignment as needed, giving you the opportunity to improve your submission before finalising it.

How to check my submissions and grade?

In the “My Grades” tab, after your assignment is graded by EmCode, the feedback can be accessed by clicking on the name of the assignment.

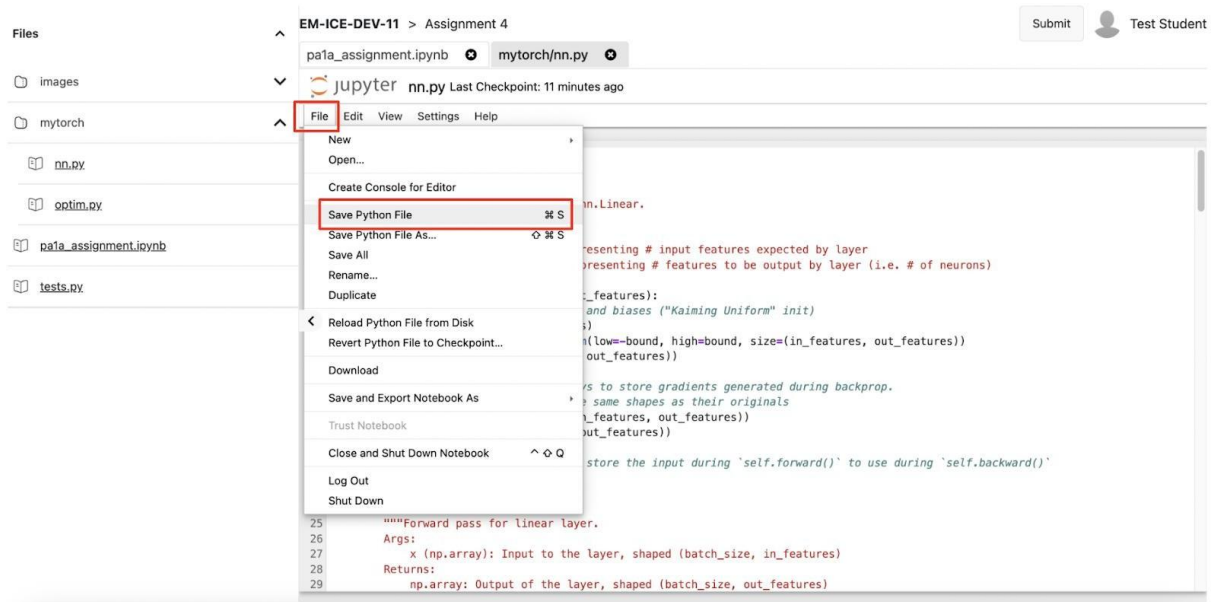
The screenshot shows the 'My Grades' tab in the course dashboard. The left sidebar contains navigation links: Home, Course Dashboard, Modules, Announcements (15), Live Sessions, People, My Grades (highlighted with a red box), Assignment Extension, Classmates, Support, and Notification Settings. The main content area is titled 'Assignments' and shows a table of assignments. The table has columns for Activity Name, Status, and Grade. The assignment 'Programming Assignment 2.1: Multilayer Perceptron (Assignment 1A) [10 hours]' is highlighted with a red box. The table also shows 'Quiz 2.1: Introduction to Learning [15 min]', 'Quiz 2.2: MLP Training [10 min]', and 'Quiz 2.3: Backpropagation [10 min]'. The status for all assignments is 'Graded'. The grade for 'Programming Assignment 2.1' is '100pts/100pts'.

The feedback will appear as follows:

The screenshot shows the feedback page for 'pa1a_assignment'. The page title is 'Assignment 1 Part A - Multi-Layer Perceptrons (MLPs)'. The page content includes a list of 25 test cells and comments. The first test cell is highlighted with a red box and labeled '1'. The 13th test cell is highlighted with a red box and labeled '2'. The 3rd test cell is highlighted with a red box and labeled '3'. The page also includes a welcome message and an introduction to the assignment.

- The section (1) at the top denotes all the cells which are autograded as part of the assignment.
- The score achieved by you out of the possible total score is mentioned for each cell (2).
- The total score for this assignment is also available at the top of the section (3).
- By selecting the name of the cell in this section, you will be redirected to the cell with the solution you have provided.

How to save a .py file?



- Click on “File” in the Menu bar
- Click on “Save Python File” to save the changes made in the .py file